Model Card: W4D1 Binary Classification Model

1. Model details

- Model: StandardScaler followed by Logistic Regression
- Task: Binary classification
- Framework: scikit-learn Pipeline
- Random seed: 42
- Original exercise: Cynaris AI/ML Internship, Week 4 Day 1

2. Intended use

This model demonstrates reproducible model training, preprocessing, evaluation, cross-validation, and explanation. It is intended for educational experimentation only. It must not be used to make automated high-impact decisions.

3. Training data

The model uses a synthetic dataset generated with scikit-learn's make_classification: 1,500 samples, 20 numerical features, 8 informative features, 4 redundant features, and an approximate 70/30 class balance. The dataset contains no real individuals' records.

4. Training procedure

The data is split into training and test sets using stratification and random seed 42. StandardScaler and LogisticRegression are combined in a Pipeline. Scaling is fitted on training data only. The original exercise uses max_iter=2000.

5. Evaluation results

- Test accuracy: 0.8067
- Test precision: 0.8511
- Test recall: 0.4396
- Test F1-score: 0.5797
- Test ROC-AUC: 0.8393
- Stratified five-fold cross-validation ROC-AUC: 0.8114 ± 0.0289

These values are from the W4D1 evaluation report. The SHAP demonstration refits the same model configuration on a stratified split; its split accuracy is recorded separately in shap_summary_metrics.json.

6. Fairness and explainability

The W4D6 exercise separately evaluates gender-related group metrics on the UCI Adult Income dataset and demonstrates AIF360 Reweighing. SHAP is used to explain the W4D1 synthetic model. Adult Income results are not a fairness audit of the W4D1 model.

7. Limitations and ethical considerations

Synthetic features do not represent real-world demographic characteristics. Consequently, demographic fairness cannot be established for the W4D1 model. The model has not been validated for hiring, lending, healthcare, policing, or other high-impact use. SHAP describes model behavior but does not prove causation

or eliminate bias.

8. Monitoring and future work

Future work should use a documented, representative dataset, assess subgroup performance where appropriate protected attributes are available, conduct independent reviews, monitor performance over time, retain human oversight, and provide a way to challenge or correct harmful decisions.